Formulation Fundamentals Booklet

Final · Lectures 15 – 17

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Each lecture is split into topics (multiple choice). The answer key is at the end of each lecture.

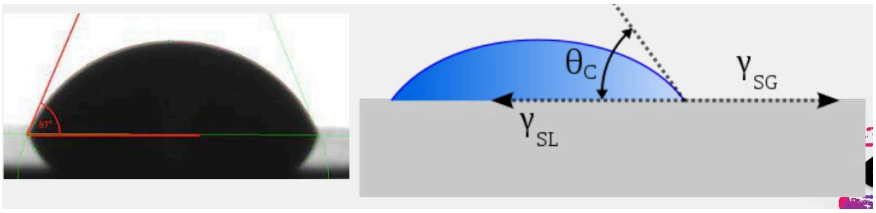
15 Solid–Liquid Interface

In this lecture

1. Interfaces & the Contact Angle 4 questions
2. Contact Angle, Cohesion & Adhesion 5 questions
3. Wetting, Spreading & Dewetting 6 questions
4. Hydrophilic & Hydrophobic Surfaces 3 questions
5. θ Is Specific for a Given System 6 questions
6. Importance of Wetting in Pharmaceutical Formulations 10 questions
7. Superhydrophobic (Lotus) Effect 6 questions

15.1 Interfaces & the Contact Angle

Q1 If a drop of liquid is placed on a flat, smooth, horizontal solid surface, the drop will:



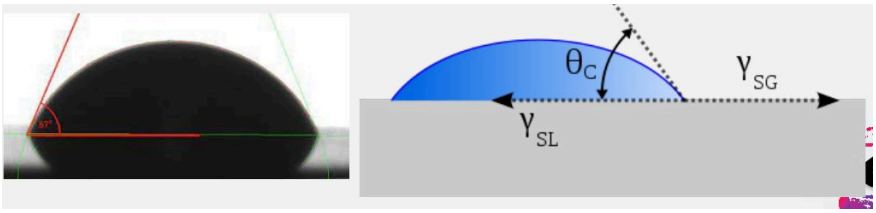
A. Exhibit a definite angle against the solid, known as the contact angle (θ, α)

B. Dissolve the solid

C. Always spread completely

D. Always form a perfect sphere

Q2 The contact angle is defined as:



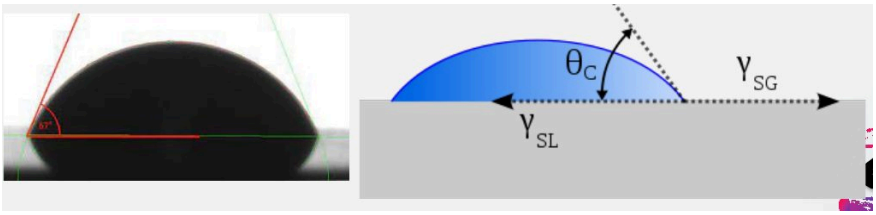
A. The angle at the top of the drop

B. The angle between two liquid surfaces

C. The angle the tangent to the liquid surface makes with the solid surface over which it spreads

D. The angle between the solid surface and the vertical

Q3 The contact angle is determined at:



A. The top of the drop

B. The point of intersection of the three interfaces: solid / liquid / gas

C. The solid–gas interface far from the drop

D. The centre of the drop

Q4 The contact angle lies in the range of:

A. 90 – 180°

B. 0 – 90°

C. 0 – 360°

D. 0 – 180°

15.2 Contact Angle, Cohesion & Adhesion

Q5 The contact angle depends on:

A. The volume of the drop only

B. The size of the container only

C. The colour of the liquid

D. The nature of the solid surface and the liquid drop

Q6 The contact angle reflects the balance between:

A. Gravity and buoyancy

B. Cohesive forces (within the solid and the liquid) and adhesive forces (between the solid and the liquid)

C. Viscosity and density

D. Temperature and pressure

Q7 A low value of θ reflects:

A. High interaction between the solid and the liquid (adhesion > cohesion)

B. Adhesion = cohesion

C. No interaction at all

D. Low interaction (adhesion < cohesion)

Q8 A high value of θ reflects:

A. Low interaction between the solid and the liquid (adhesion < cohesion)

B. Adhesion = cohesion

C. Complete spreading

D. High interaction (adhesion > cohesion)

Q9 θ for pure water on pure silver is 90°. This means:

A. Adhesion < cohesion

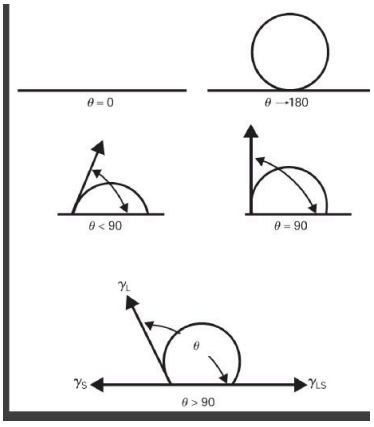
B. Complete spreading

C. Adhesion > cohesion

D. Adhesion = cohesion

15.3 Wetting, Spreading & Dewetting

Q10 If the attractive forces between water and the solid surface exceed the cohesive forces in water:



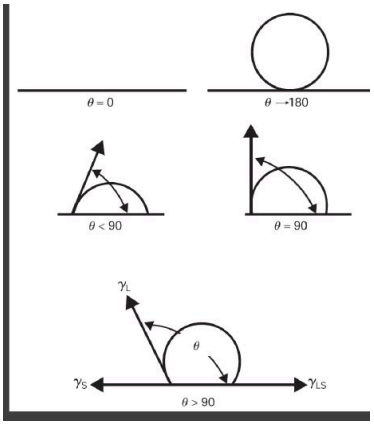
A. Large contact angle (θ > 90°) → incomplete wetting

B. Small contact angle (θ < 90°) → good wetting, e.g. water on glass

C. θ = 90° always

D. θ = 180° → non-wetting

Q11 If the attractive forces between the liquid and the solid are less than the cohesive forces in the liquid:



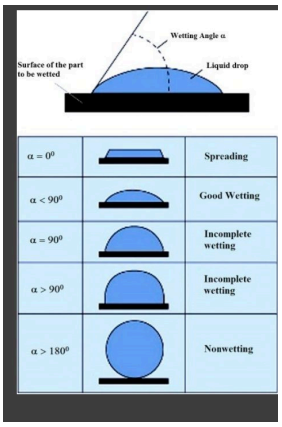
A. The drop dissolves the solid

B. Small contact angle (θ < 90°) → good wetting

C. θ = 0° → spreading

D. Large contact angle (θ > 90°) → incomplete wetting, e.g. water on Teflon

Q12 Spreading occurs when:



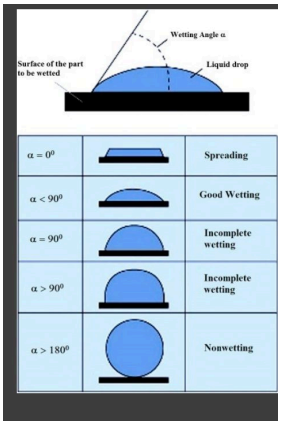
A. θ = 90°

B. θ > 90°

C. θ = 180° → minimal interaction

D. θ = 0° → maximum adhesive forces

Q13 Non-wetting / dewetting takes place at:



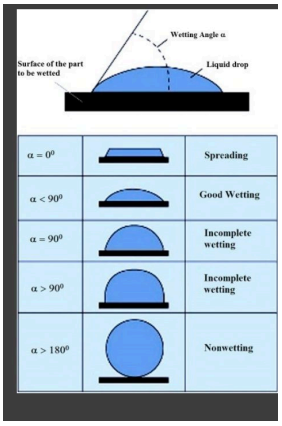
A. θ = 180° → minimal interaction between the solid and liquid

B. θ = 45°

C. θ < 90°

D. θ = 0°

Q14 From the table: a contact angle of α < 90° means ___, while α = 90° and α > 90° mean ___.



A. Good wetting; incomplete wetting

B. Spreading; non-wetting

C. Non-wetting; spreading

D. Incomplete wetting; good wetting

Q15 By knowing the contact angle we can differentiate between:

A. Melting and boiling

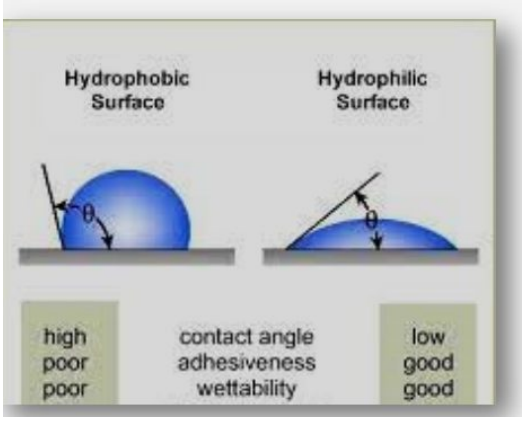
B. Wetting, dewetting and spreading

C. Acids and bases

D. Solutions and suspensions only

15.4 Hydrophilic & Hydrophobic Surfaces

Q16 A surface with a contact angle < 90° with water is:



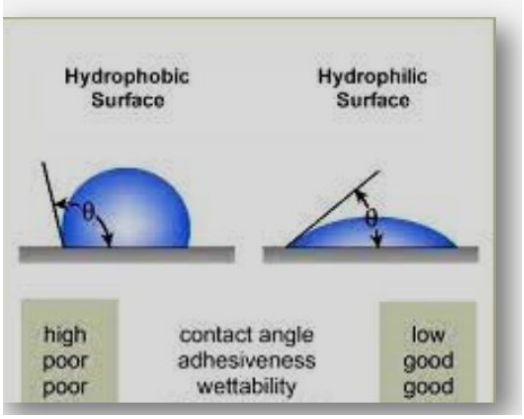
A. Lipophilic

B. Hydrophilic

C. Superhydrophobic

D. Hydrophobic

Q17 A surface with a contact angle > 90° with water is:



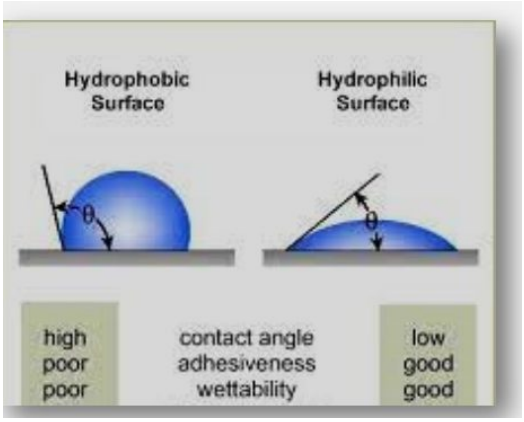
A. Spreading

B. Hydrophilic

C. Hydrophobic

D. Completely wetted

Q18 Compared with a hydrophilic surface, a hydrophobic surface has:



A. Low contact angle and poor wettability

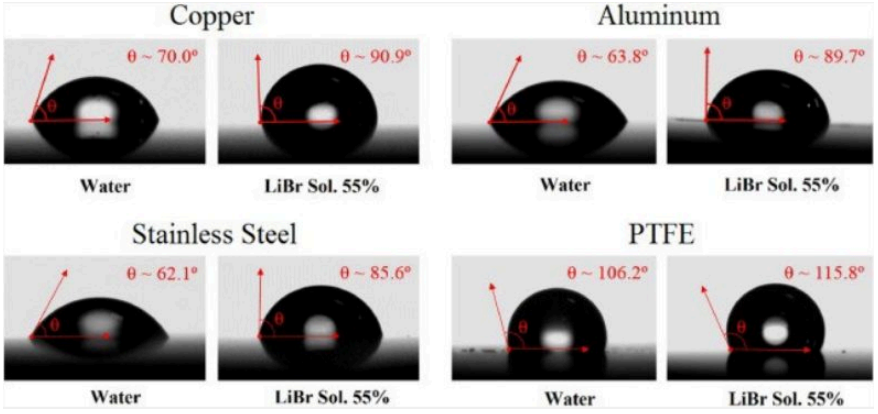
B. High contact angle and good wettability

C. High contact angle, poor adhesiveness and poor wettability

D. Low contact angle, good adhesiveness and good wettability

15.5 θ Is Specific for a Given System

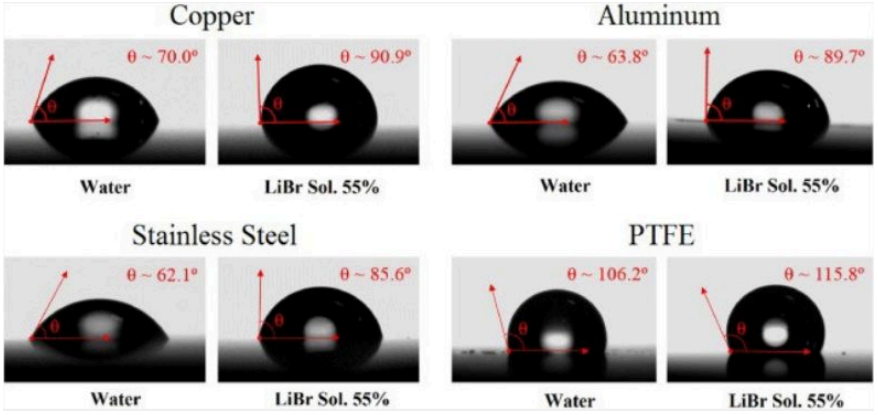
Q19 θ water–copper (70.0°) < θ LiBr–copper (90.9°). This means that the adhesive forces between water and copper are stronger than those between LiBr and copper.



A. True

B. False

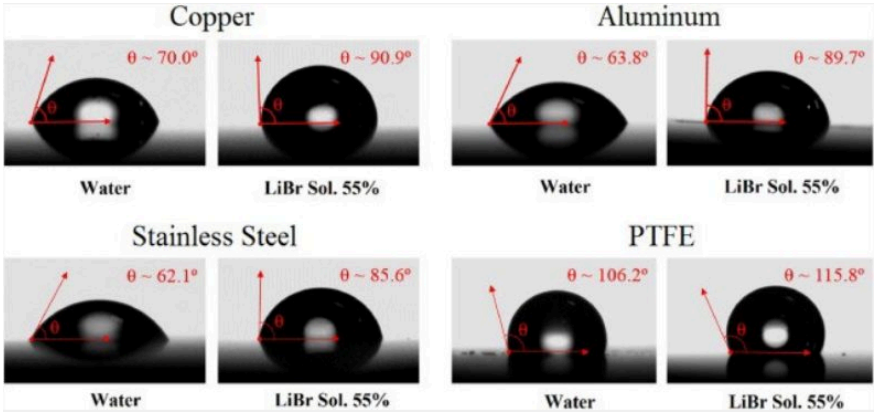
Q20 The addition of LiBr to water (LiBr solution 55%) increases its interaction with the aluminum surface.



A. True

B. False

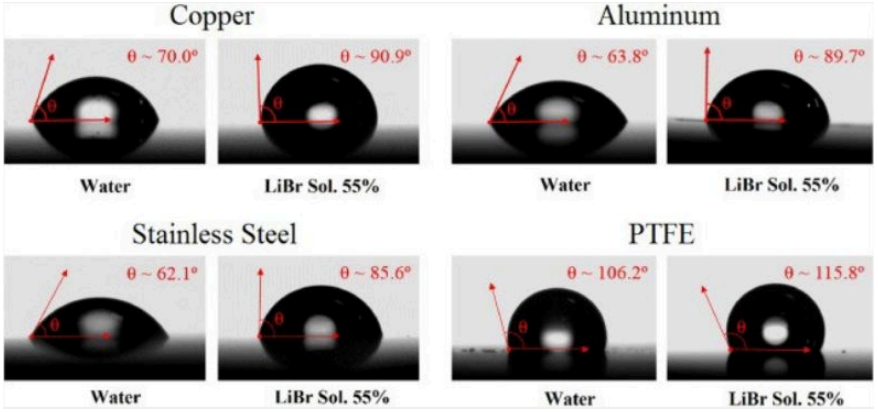
Q21 The stainless steel surface is better wetted with water compared to PTFE.



A. True

B. False

Q22 Which system ensures the maximum solid–liquid interaction?



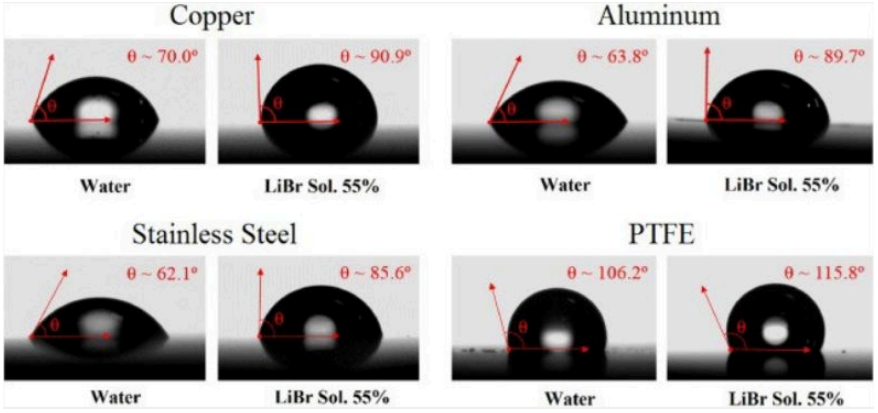
A. LiBr 55% on PTFE (θ ≈ 115.8°)

B. LiBr 55% on copper (θ ≈ 90.9°)

C. Water on stainless steel (θ ≈ 62.1°)

D. Water on PTFE (θ ≈ 106.2°)

Q23 Which system shows the minimum solid–liquid interaction?



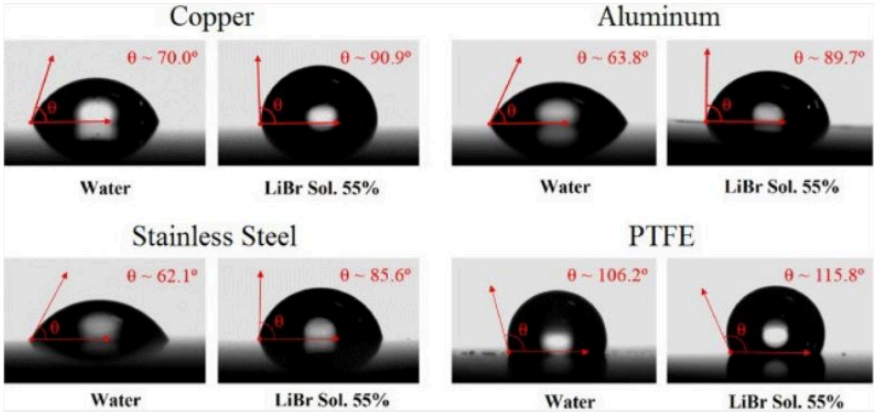
A. LiBr 55% on PTFE (θ ≈ 115.8°)

B. Water on aluminum (θ ≈ 63.8°)

C. Water on stainless steel (θ ≈ 62.1°)

D. Water on copper (θ ≈ 70.0°)

Q24 Which surface is hydrophobic in this figure?



A. PTFE (θ with water ≈ 106.2° > 90°)

B. Stainless steel

C. Copper

D. Aluminum

15.6 Importance of Wetting in Pharmaceutical Formulations

Q25 Why should suspended solid particles have a hydrophilic coating (wetting agent)?

A. To make the suspension hydrophobic

B. To increase particle size

C. To make the particles dissolve completely

D. To ensure good wetting and dispersion and avoid particle agglomeration

Q26 A suspension is composed of:

A. A gas dispersed in a liquid

B. Two immiscible liquids

C. Insoluble solid particles suspended in water

D. A drug completely dissolved in water

Q27 What dictates how fast a tablet disintegrates?

A. The amount of air in the bottle

B. The tablet's shape only

C. The speed at which water or gastric fluid penetrates the compressed tablet matrix

D. The colour of the tablet

Q28 A poorly-wetted tablet leads to:

A. Poor water penetration into the tablet core → slow disintegration

B. Fast disintegration

C. Faster dissolution

D. Better swelling

Q29 Good wettability of the disintegrated particles leads to:

A. No drug release

B. Lower dissolution rate

C. Higher chance for the drug to dissolve → high dissolution / release rate

D. Agglomeration

Q30 To prepare immediate-release tablets, ___ excipients should be used; for sustained-release tablets, ___ excipients.

A. Hydrophobic; hydrophobic

B. Hydrophobic; hydrophilic

C. Hydrophilic; hydrophilic

D. Hydrophilic; hydrophobic

Q31 In tablets that remain intact (no disintegration), the drug is released:

A. Never

B. Immediately

C. Only after the tablet melts

D. Slowly, after swelling of the intact tablet (water absorption and gel-layer formation)

Q32 Swelling of a tablet means:

A. Breaking into fine particles

B. An increase in tablet dimension following water absorption and formation of a gel layer

C. Melting of the tablet

D. Loss of the coating

Q33 In wet granulation, the liquid binder solution should:

A. Dissolve the granules

B. Not touch the powder

C. Have a high contact angle with the powder

D. Ensure good wetting of the powder (low θ) to form granules

Q34 In tablet coating, the coating solution should:

A. Have θ = 180° with the tablet

B. Be able to wet the tablet surface to ensure good coating

C. Repel the tablet surface

D. Dissolve the tablet core

15.7 Superhydrophobic (Lotus) Effect

Q35 The “lotus effect” refers to:



A. Complete spreading of water

B. The fast dissolution of tablets

C. The yellow colour of lotus flowers

D. The very high water repellence of lotus leaves (superhydrophobic effect)

Q36 Super-hydrophobic surfaces show contact angles up to about:



A. 90°

B. 60° — most of the drop is in contact

C. 160° — only 2–3% of a drop's surface is in contact

D. 0°

Q37 The superhydrophobic effect is also called:

A. The dissolution effect

B. The wetting effect

C. The self-cleaning effect

D. The spreading effect

Q38 Why do superhydrophobic surfaces stay clean?

A. Dirt dissolves in the surface

B. Dirt particles have minor contact with the surface, and their adhesion to water is much higher than to the surface

C. Water spreads and soaks the dirt into the surface

D. The surface is hydrophilic

Q39 Which is a general (technical) application of the superhydrophobic effect?

A. Water-proof coatings, paints, roof tiles, fabrics and cars

B. Hydrophilic wetting agents in suspensions

C. Tablet disintegrants

D. Wet granulation binders

Q40 Which is a pharmaceutical application of the superhydrophobic effect?

A. Oral rehydration solutions

B. Plasters, blisters & containers, water-resistant cosmetics & sunscreens, nanoparticles for drug delivery

C. Wetting agents in suspensions

D. Immediate-release tablets

Answers · Lecture 15

Q1 A	**Q9** D	**Q17** C	**Q25** D	**Q33** D
Q2 C	**Q10** B	**Q18** C	**Q26** C	**Q34** B
Q3 B	**Q11** D	**Q19** A	**Q27** C	**Q35** D
Q4 D	**Q12** D	**Q20** B	**Q28** A	**Q36** C
Q5 D	**Q13** A	**Q21** A	**Q29** C	**Q37** C
Q6 B	**Q14** A	**Q22** C	**Q30** D	**Q38** B
Q7 A	**Q15** B	**Q23** A	**Q31** D	**Q39** A
Q8 A	**Q16** B	**Q24** A	**Q32** B	**Q40** B

16 Surface-Active Agents & CMC

In this lecture

1. Surfactants & Surface Tension 9 questions
2. Micelles & CMC 11 questions

16.1 Surfactants & Surface Tension

Q1 Surface active agents (SAA, surfactants) are described as:

A. Amphiphilic molecules
B. Purely lipophilic molecules
C. Purely hydrophilic molecules
D. Inorganic salts

Q2 When SAA is added to water, the molecules arrange at the surface so that:

A. The tails dip into the water and the heads point to the air
B. The polar heads associate with water and the lipophilic tails point upwards to the air
C. All molecules dissolve evenly in the bulk
D. The molecules lie flat with no orientation

Q3 Which part of a surfactant molecule has affinity to polar, hydrophilic liquids?

A. The hydrocarbon chain
B. The polar (hydrophilic) head
C. Both ends equally
D. The lipophilic tail

Q4 When surfactant molecules adsorb at the water surface, they:

A. Push all water molecules into the air
B. React chemically with water
C. Replace some of the water molecules in the surface
D. Stay only in the bulk of the liquid

Q5 On slide 4, surface tension is described as the:

A. Pressure of the air above the liquid
B. Attraction between water and the container
C. Contraction force that results from cohesive forces between water molecules
D. Force of gravity on the liquid

Q6 Adding SAA to water affects its surface tension by:

A. Not changing it
B. Reducing it
C. Increasing it
D. Doubling it

Q7 Why does SAA reduce the surface tension of water?

A. Surfactant molecules increase hydrogen bonding at the surface

B. Attraction between surfactant and water molecules is less than between two water molecules, so cohesive forces at the surface decrease

C. Surfactants evaporate the surface water molecules

D. Surfactants raise the temperature of the water

Q8 When oil is added to a water/SAA system, the SAA molecules orient so that:

A. Polar heads associate with water and lipophilic tails associate with the oil

B. Polar heads associate with the oil and tails with water

C. All SAA moves into the oil phase

D. All SAA stays in the water bulk

Q9 SAA at an oil–water interface will:

A. Make oil and water completely immiscible

B. Have no effect on interfacial tension

C. Reduce the interfacial tension between oil and water

D. Increase the interfacial tension

16.2 Micelles & CMC

Q10 Micelles in water are assemblies of SAA in which:

A. Lipophilic tails face water and heads are hidden inside

B. Hydrophilic heads face water and lipophilic tails are hidden inside

C. Only tails are present

D. Heads and tails are randomly mixed

Q11 As more SAA is added to water, the water surface becomes saturated. This means:

A. Complete surface coverage — excess SAA molecules then go into the liquid bulk

B. The SAA forms crystals on the surface

C. The surface tension rises back to that of pure water

D. All the SAA evaporates

Q12 SAA molecules aggregate into micelles in water in order to:

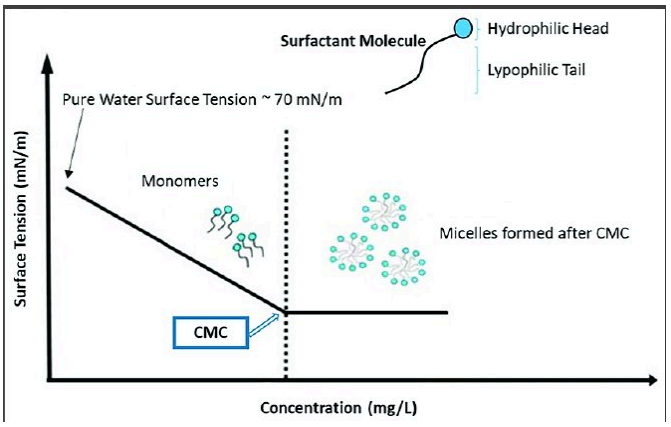
A. Expose their tails to the water

B. Bind to oil droplets only

C. Increase the surface tension

D. Hide their hydrophobic tails from the hydrophilic water

Q13 Below the CMC, the surfactant in solution exists mainly as:



A. Gel

B. Micelles

C. Inverted micelles

D. Monomers (single molecules)

Q14 Adding SAA above the CMC continues to lower the surface tension of water.

A. True

B. False

Q15 In inverted micelles, the lipophilic tails are directed outwards into the oil.

A. True

B. False

Q16 The critical micelle concentration (CMC) is:

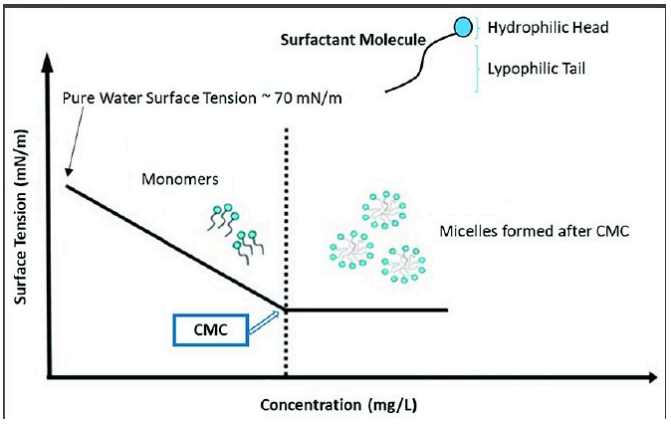
A. The concentration at which SAA starts to adsorb at the surface

B. The maximum solubility of SAA in water

C. The concentration of SAA at which micelles start to form

D. The concentration at which surface tension equals zero

Q17 According to the graph, adding more SAA above the CMC will:



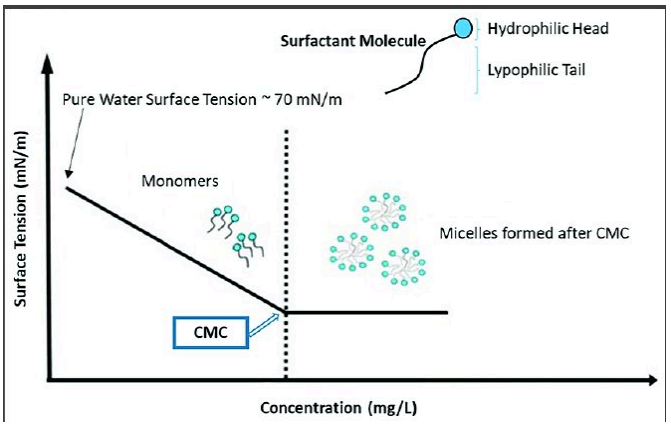
A. Increase the number of micelles without further reduction in surface tension

B. Increase surface tension back to ~70 mN/m

C. Keep reducing surface tension sharply

D. Destroy the existing micelles

Q18 On the CMC graph, the surface tension of pure water (before any SAA) is about:



A. 70 mN/m

B. 180 mN/m

C. 7 mN/m

D. 20 mN/m

Q19 Which statement about surfactant at and above the CMC is TRUE?

A. Micelles only form at the surface

B. Surface and bulk molecules never exchange

C. All surfactant molecules leave the surface

D. There is an equilibrium between surfactant molecules at the surface and those in the bulk

Q20 When SAA is added to oil or a lipophilic solution, it forms:

A. Crystals

B. A monolayer only

C. Normal micelles: heads outside, tails inside

D. Inverted micelles: hydrophilic heads protected inside, lipophilic tails directed outwards

Answers · Lecture 16

Q1 A	**Q5** C	**Q9** C	**Q13** D	**Q17** A
Q2 B	**Q6** B	**Q10** B	**Q14** B	**Q18** A
Q3 B	**Q7** B	**Q11** A	**Q15** A	**Q19** D
Q4 C	**Q8** A	**Q12** D	**Q16** C	**Q20** D

17 HLB & Types of SAA

In this lecture

1. Hydrophilic–Lipophilic Balance (HLB) 4 questions
2. HLB & Applications of SAA 8 questions
3. Types of SAA Based on Their Charge 7 questions
4. Poloxamers 5 questions
5. Spans & Tweens 7 questions
6. Biological Surfactants 4 questions

17.1 Hydrophilic–Lipophilic Balance (HLB)

Q1 Whether a SAA is predominantly hydrophilic, lipophilic or balanced depends on:

A. The size of the container

B. Its colour

C. The number and nature of its polar and non-polar groups

D. The temperature of the room only

Q2 Ethyl alcohol, amyl alcohol and cetyl alcohol all have one OH group. Which is the most lipophilic?

A. Cetyl alcohol (longest hydrocarbon chain)

B. Ethyl alcohol (shortest chain)

C. They are equally lipophilic

D. Amyl alcohol

Q3 HLB is:

A. A scale of 0–14 for acidity

B. The concentration at which micelles form

C. An arbitrary scale (0–20) devised by Griffin to measure the hydrophilicity/lipophilicity of a SAA

D. A measure of the charge of a SAA

Q4 A SAA with an HLB lower than 10 is:

A. More lipophilic

B. Always cationic

C. More hydrophilic

D. Always a detergent

17.2 HLB & Applications of SAA

Q5 Extremely lipophilic SAA (HLB < 3) are commonly used as:

A. Detergents

B. Antifoaming agents

C. O/W emulsifiers

D. Solubilizers

Q6 Lipophilic SAA with HLB 3–8 can be used as:

A. Emulsifiers to prepare W/O emulsions

B. Emulsifiers to prepare O/W emulsions

C. Detergents

D. Solubilizers

Q7 Hydrophilic SAA with HLB 8–16 can be used as emulsifiers for:

A. Inverted micelles only

B. O/W emulsions

C. Antifoaming

D. W/O emulsions

Q8 SAA with a balanced hydrophilic/lipophilic character (HLB 7–9) tend to be used as:

A. Wetting agents

B. Solubilizers

C. Antifoaming agents

D. Detergents

Q9 Highly hydrophilic SAA (HLB > 13) are used as:

A. Antifoaming agents

B. Detergents and solubilizers (above CMC)

C. W/O emulsifiers

D. Wetting agents only

Q10 Slide 13 asks “So what does the foam do?” In a shampoo, the foam:

A. Is the part that removes the dirt by itself

B. Increases the surface tension of water

C. Creates lather and helps spread the shampoo, while the surfactants lift oil, dirt and residue

D. Kills bacteria on the scalp

Q11 In micellar solubilization, the SAA concentration should be:

A. Equal to half the CMC

B. Below the CMC

C. Above the CMC

D. Exactly zero

Q12 What is the application of antifoaming agents in pharmaceutical manufacture?

A. To act as antibacterials

B. To produce more foam in shampoos

C. To increase the surface tension of every product to 100 mN/m

D. To prevent or break foam formed during processing (e.g. mixing, stirring, filling), so manufacturing proceeds smoothly

17.3 Types of SAA Based on Their Charge

Q13 SAA are classified according to the charge on their:

A. Micelle core

B. Counter-ion only

C. Hydrophilic head

D. Lipophilic tail

Q14 Sodium lauryl sulphate is an example of which type of SAA?

A. Non-ionic

B. Amphoteric

C. Anionic

D. Cationic

Q15 Anionic SAA are commonly used as detergents because:

A. They have no charge

B. Their charge changes with pH

C. They kill bacteria by binding to the cell membrane

D. Their negatively charged head gives high washing power, they are inexpensive, effective and produce abundant foam

Q16 Benzalkonium chloride and cetyl pyridinium chloride are:

A. Amphoteric SAA used in cosmetics

B. Non-ionic SAA used as emulsifiers

C. Cationic SAA (quaternary ammonium compounds) used as antibacterials

D. Anionic SAA used as detergents

Q17 Cationic SAA are antibacterial because:

A. Their positive head interacts with the negatively charged bacterial membrane and the tail penetrates the inner membrane, disrupting it

B. They produce a lot of foam

C. They have no charge so they enter cells freely

D. They reduce the pH of the bacteria

Q18 N-dodecyl alanine and betaines are amphoteric SAA. Their main feature is:

A. They are always negatively charged

B. They are used as antifoaming agents

C. They have no charge at any pH

D. Their charge changes according to pH changes; used in cosmetics

Q19 Spans and Tweens belong to which class of SAA?

A. Anionic

B. Non-ionic

C. Amphoteric

D. Cationic

17.4 Poloxamers

Q20 Poloxamers (Pluronics) consist of:

A. Three blocks: two hydrophilic blocks and a central hydrophobic block

B. A sorbitan ester of a fatty acid

C. One hydrophilic head and two lipophilic tails

D. Two hydrophobic blocks and a central hydrophilic block

Q21 In the poloxamer structure, increasing “x” (the hydrophilic outer parts) makes the poloxamer:

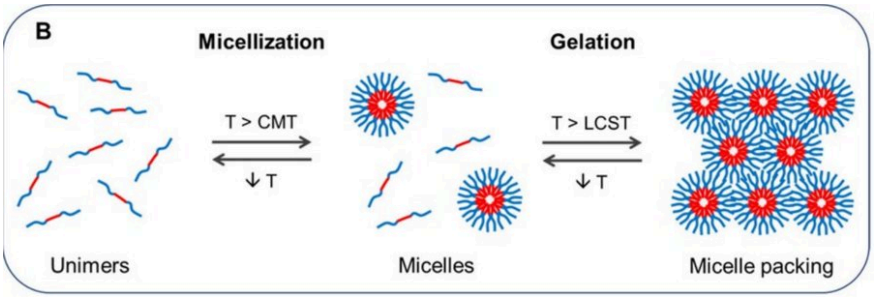
A. Cationic

B. Gel at a lower temperature

C. More insoluble in water

D. More soluble in water

Q22 Poloxamers are suitable for applications requiring thermal gelation because:



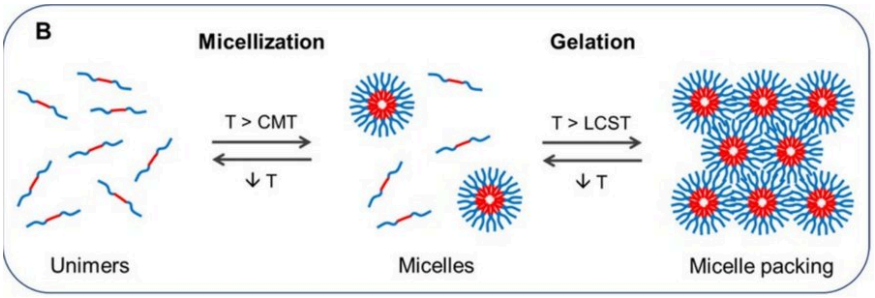
A. They are in a sol state at low temperature/concentration and turn into a gel at higher temperature/concentration

B. Their charge changes with pH

C. They are always solid

D. They melt into a liquid when heated

Q23 In the figure, unimers form micelles when T > CMT, then the micelles pack into a gel when:



A. Water evaporates completely

B. The pH is lowered

C. T falls below CMT

D. T > LCST

Q24 Which is NOT listed as a pharmaceutical application of poloxamers?

A. Antifoaming agents

B. Wound dressing

C. Eye drops

D. Injectable hydrogels

17.5 Spans & Tweens

Q25 Spans are:

A. Polyoxyethylene sorbitan esters, more soluble in water, used for O/W emulsions

B. Sorbitan esters of fatty acids with a small hydrophilic head, more soluble in oil, used for W/O emulsions (low HLB)

C. Quaternary ammonium compounds

D. Three-block copolymers

Q26 Tweens differ from Spans because Tweens are:

A. Free of any fatty acid

B. Modified by the addition of polyoxyethylene chains, making them more water soluble (high HLB, O/W)

C. Made from a different sugar than sorbitan

D. Cationic

Q27 Span 20, Span 60 and Span 80 are made with which fatty acids, respectively?

A. Stearic, oleic, lauric

B. Oleic, lauric, stearic

C. Lauric, oleic, stearic

D. Lauric, stearic, oleic

Q28 Different Tweens (e.g. Tween 20 and Tween 80) have:

A. Different hydrophilic heads but the same tail

B. No hydrophilic head

C. The same hydrophilic head but different lipophilic tails

D. The same head and the same tail

Q29 According to the table, Span 80 (sorbitan oleate) has HLB 4.3. It is best used as:

Names of surfactants	*HLB*
Sorbitan laurate (Span 20)	8.6
Sorbitan palmitate (Span 40)	6.7
Sorbitan stearate (Span 60)	4.7
Sorbitan oleate (Span 80)	4.3
Sorbitan trioleate (Span 85)	1.8
Polyoxyethylene sorbitan laurate (Tween 20)	16.7
Polyoxyethylene sorbitan palmitate (Tween 40)	15.6
Polyoxyethylene sorbitan stearate (Tween 60)	14.9
Polyoxyethylene sorbitan oleate (Tween 80)	15.0
Polyoxyethylene sorbitan trioleate (Tween 85)	11.0

A. A W/O emulsifier

B. A solubilizer

C. A detergent

D. An O/W emulsifier

Q30 According to the table, which surfactant has the lowest HLB and would act as an antifoaming agent?

Names of surfactants	*HLB*
Sorbitan laurate (Span 20)	8.6
Sorbitan palmitate (Span 40)	6.7
Sorbitan stearate (Span 60)	4.7
Sorbitan oleate (Span 80)	4.3
Sorbitan trioleate (Span 85)	1.8
Polyoxyethylene sorbitan laurate (Tween 20)	16.7
Polyoxyethylene sorbitan palmitate (Tween 40)	15.6
Polyoxyethylene sorbitan stearate (Tween 60)	14.9
Polyoxyethylene sorbitan oleate (Tween 80)	15.0
Polyoxyethylene sorbitan trioleate (Tween 85)	11.0

A. Polyoxyethylene sorbitan trioleate (Tween 85)

B. Sorbitan trioleate (Span 85)

C. Sorbitan laurate (Span 20)

D. Sorbitan stearate (Span 60)

Q31 Tween 20 (lauric) has HLB 16.7 while Tween 80 (oleic) has HLB 15.0. This shows that:

Names of surfactants	*HLB*
Sorbitan laurate (Span 20)	8.6
Sorbitan palmitate (Span 40)	6.7
Sorbitan stearate (Span 60)	4.7
Sorbitan oleate (Span 80)	4.3
Sorbitan trioleate (Span 85)	1.8
Polyoxyethylene sorbitan laurate (Tween 20)	16.7
Polyoxyethylene sorbitan palmitate (Tween 40)	15.6
Polyoxyethylene sorbitan stearate (Tween 60)	14.9
Polyoxyethylene sorbitan oleate (Tween 80)	15.0
Polyoxyethylene sorbitan trioleate (Tween 85)	11.0

A. All Tweens have the same HLB

B. The choice of fatty acid affects the surfactant's HLB value

C. Oleic acid is more hydrophilic than lauric acid

D. HLB depends only on the head group

17.6 Biological Surfactants

Q32 The cell membrane acts as a biological SAA because it is composed of:

A. A single layer of proteins

B. Cholesterol only

C. Polyoxyethylene chains

D. A phospholipid bilayer; each phospholipid has a hydrophilic head and 2 lipophilic tails

Q33 How does the phospholipid bilayer of the cell membrane affect drug absorption/penetration?

A. It allows all drugs to pass equally

B. It blocks all lipophilic drugs

C. Only highly ionized drugs can cross it

D. Its lipophilic core favours penetration of drugs with enough lipophilicity, while highly polar/ionized drugs cross poorly

Q34 In the cell membrane, cholesterol is positioned so that:

A. It replaces all phospholipid heads

B. Its hydroxyl group is buried in the centre of the bilayer

C. Its hydrophobic region is embedded alongside the fatty-acid chains and its hydroxyl group points towards water

D. It floats freely outside the membrane

Q35 Lung surfactant is a lipoprotein complex at the alveolar air–liquid interface. Its main role is to:

A. Kill bacteria in the lungs

B. Reduce surface tension so the lungs do not collapse when we breathe out

C. Increase surface tension to keep alveoli closed

D. Form inverted micelles in the blood

Answers · Lecture 17

Q1 C	**Q8** A	**Q15** D	**Q22** A	**Q29** A
Q2 A	**Q9** B	**Q16** C	**Q23** D	**Q30** B
Q3 C	**Q10** C	**Q17** A	**Q24** A	**Q31** B
Q4 A	**Q11** C	**Q18** D	**Q25** B	**Q32** D
Q5 B	**Q12** D	**Q19** B	**Q26** B	**Q33** D
Q6 A	**Q13** C	**Q20** A	**Q27** D	**Q34** C
Q7 B	**Q14** C	**Q21** D	**Q28** C	**Q35** B